RAKESH CHOWDARY POLINA

rpolina@okstate.edu | https://www.linkedin.com/in/rakesh-chowdary-polina | 405-894-3989

PROFILE

Experienced Software Engineer with a proven track record in developing scalable microservices, implementing robust facial recognition pipelines, and delivering cloud-native solutions. Skilled in Python, Java, Spring Boot, Flask, React.js, and AWS, with expertise in containerization, CI/CD, and Agile methodologies. Adept at optimizing performance, enhancing user engagement, and driving operational efficiency in production environments.

EDUCATION

Master of Science in Computer Science **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

SKILLS

Languages:	Java, JavaScript, Python, SQL, C, C++, HTML, CSS, Swift
Frameworks:	React JS, Angular JS, Spring boot, Node JS, Hibernate, Spring MVC, React Native
Databases:	MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra
Web Technologies:	RESTful API, Spring JDBC, JPA, OAuth 2, jQuery, JSP, JMS, RabbitMQ
Web Services:	REST, SOAP
Tools/Platforms:	Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger
AWS Cloud Services:	EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront
Build Tools:	Maven, Gradle

WORK EXPERIENCE

Research Data Analyst, Oklahoma State University **May 2023 – May 2024**

- Engineered a facial recognition pipeline using Python and InsightFace, performing quantitative analysis on 182,000+ institutional images to achieve 93.6% precision. Integrated Ellucian Banner ERP data (student/faculty records) to validate identity metadata, ensuring alignment with institutional databases.
- Improved data reliability by implementing DBSCAN clustering to detect and resolve anomalies in image archives, enhancing research accuracy by 22%. Automated workflows using SQL and Python to reconcile discrepancies with Ellucian ERP datasets.
- Developed a Flask-based internal tool with React.js for streamlined image labeling and annotation, reducing manual effort by 35%. Configured JSON/XML outputs for compatibility with downstream systems, including Ellucian-powered reporting dashboards.
- Collaborated with institutional IT to troubleshoot integration issues between analytics tools and the Ellucian environment, resolving 15+ API/data synchronization errors through log analysis and Bash scripting.

Software Engineer, Ericsson, India **July 2019 – July 2022**

- Modernized legacy monolithic applications (including PHP/Java EE modules) to cloud-native microservices using Spring Boot and RESTful APIs, improving scalability to handle 50% more concurrent users. Retrofitted legacy PHP workflows into containerized Spring Boot services for better cloud compatibility.
- Migrated on-prem solutions to AWS (EC2, RDS, S3) by refactoring Java services, integrating Elastic Load Balancer and Auto Scaling for high availability for critical Oracle-based billing systems.
- Implemented event-driven architecture with Apache Kafka for asynchronous communication, reducing system coupling and enhancing fault tolerance.
- Developed full-stack solutions using Java/Spring Boot for back-end services and React for dynamic front-end interfaces. Designed REST/GraphQL APIs with Spring WebFlux for reactive, non-blocking performance, reducing latency by 40%.
- Utilized Spring Data JPA and PostgreSQL for efficient data modeling, optimizing complex SQL queries to cut query execution time by 25%. Integrated Oracle BI Publisher for automated financial and operational reporting.
- Built CI/CD pipelines with Jenkins and Git, automating deployments to AWS EKS clusters. Integrated SonarQube for security scanning and code quality checks, reducing vulnerabilities by 30%.
- Containerized applications using Docker and orchestrated with Kubernetes, cutting deployment time by 50% and enabling zero-downtime releases.
- Conducted end-to-end testing with JUnit/Mockito, achieving 95% test coverage, and used Postman for API contract validation.
- Collaborated with Agile teams using Jira/Confluence for sprint planning, backlog refinement, and documentation.
- Partnered with DevOps to monitor production systems via New Relic and Splunk, resolving critical issues 20% faster. Delivered bi-weekly releases aligned with sprint goals, improving feature delivery consistency by 25%.
- Designed PostgreSQL schemas for high-volume transactional systems, optimizing indexes and partitioning strategies. Integrated React front-end components with Spring MVC for server-side rendering, enhancing UI load times by 30%.
- Troubleshot live production issues using SLF4J logging and AWS CloudWatch, reducing downtime by 30%. Refactored legacy Java EE codebases to Spring Boot, eliminating deprecated dependencies and improving maintainability.